BIOS 755: Growth Curves and Confounding

Alexander McLain

Modeling the Mean: Two Key Questions

- In longitudinal linear models, our first job is to specify the **mean response trajectory**.
- Doing so boils down to *two* fundamental questions:
  1. **Which** covariates should appear in the model?
  2. **How** should each chosen covariate be incorporated?
     - Baseline vs. time-varying versions,
     - Transformations, interactions, random slopes, etc.
- **Roadmap for this week:**
  - We'll begin with how to include covariates (coding and structure).
  - Then we'll focus on which covariates to include, i.e. confounder selection (upcoming slides).

Including Covariates: Binary $\rightarrow$ Continuous

- **Binary predictors (easy case)**
  - If you decide to put a binary predictor in your model, there is no question of how to include it.
  - Code as 0/1 indicator.
  - Coefficient = mean difference between groups.
  - Interaction with time? If yes, you'll get two effects of time: one for the referent '0' group, and one for the difference.

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  - Interaction with time? If yes, you'll get two effects of time: one for the referent '0' group, and one for the difference.
- **Nominal predictors (e.g., race/Ethnicity, type of illness, ...)**
  - Create dummy variables.
  - One category will be the referent group, for the remaining, we'll get a coefficient for the mean difference between the specific category and the referent group.
  - For example, if we use EA as the referent group, we'll get estimates for the mean difference between EA and AA, EA and Hispanics, etc.

Including Covariates: Binary → Continuous

- **Ordinal / count (Education level, Pain severity, cigarettes smoked per day)**
  - These will be treated mostly like continuous predictors (below), but dummy coding is more common.
- **Continuous predictors (time, BMI, stress, MVPA, age, etc.)**
  - There are two options:
    - Profile Analysis (i.e., dummy coding)
    - Parametric Curves (Linear and Nonlinear Trend)
  - We'll discuss profile analysis below for **Time**, we'll discuss parametric curves next time.
  - While we focus on time as a continuous covariate, these issues hold for **any continuous exposures:** BMI, stress, BP, age, etc.

Treatment of Lead-Exposed Chidren (TLC) Trial

- Facts of the trial:
  - Randomized trial,
  - 100 children randomized to placebo or Succimer,
  - measures of blood lead level at baseline, 1, 4, and 6 weeks.
- For randomized trials, there is no need to adjust for confounders (e.g., age, race/ethnicity, SES).

The Model

- The main goal of this study is to determine if the outcomes are different between the treatment and control groups.
- Since the treatment is randomized, the groups should be the same at baseline.
- Over the study, the only difference between the groups is the treatment.
- If the impact of time is different between the groups, this is a strong indication that the drug is having an impact on the outcome.
- A commonly used model is:

$$Y_{ij} = \beta_0 + \beta_1 Group_i + \beta_2 Time_j + \beta_3 Time_j * Group_i + \epsilon_{ij}$$

- **How to include time in the model?**

Three-Step Process for Including a Continuous Covariate

1. **Explore Visually** – Plot the outcome against the covariate (spaghetti + group means) to reveal shape.
2. **Generate Functional Forms** – Propose candidate representations (linear, polynomial, spline, etc.).
3. **Compare Select** – Fit each form, evaluate model fit, and choose the best-performing specification.

We'll walk through each step on the next slides.

Step 1 – Visual Exploration

1. **Plot the Raw Data**
    - Create a *spaghetti plot* of individual trajectories.
    - *X-axis:* continuous covariate.
    - *Y-axis:* outcome mean (over time or at a specific time point).
2. **Add Group Structure (if applicable)**
    - Color or facet by exposure/treatment group, sex, cohort, etc.
    - Overlay group-specific means or LOESS curves to spot non-linearity.
3. **Goal**
    - Visually assess shape (linear? curved? threshold?) before formal modeling.

Steps 2 & 3 – Choosing the Best Functional Form

2. **Generate Candidate Specifications**
    - Profile analysis (categorical bins)
    - Linear term (βX)
    - Transformation ($\log(X)$, $\sqrt{X}$)
    - Quadratic or cubic polynomial ($\beta_1 X + \beta_2 X^2$)
    - Piecewise linear (knots) or restricted cubic splines
3. **Compare and Select**
    - Fit each candidate in the longitudinal model.
    - Evaluate with AIC/BIC.
    - **Choose the form with the best fit** that is also interpretable and consistent with subject-matter knowledge.

Tip: Document your decision tree—why each form was tried and why the final form was selected—for transparency and reproducibility.

Profile Analysis

- First, we'll explore including time using dummy coding.
- With dummy coding, we choose one time point as the baseline time point.
- The mean for the rest of the time points is modeled relative to the baseline time point.
- Profile analysis is **always** an option.

Profile coding example

- **Three time points** and **two groups** with t_{i1} being the baseline time.
- We'll have two "time coefficients" and on "group coefficient"

$$T_{ij2} = \begin{cases} 1 & \text{if j=2} \\ 0 & \text{otherwise} \end{cases} \qquad T_{ij3} = \begin{cases} 1 & \text{if j=3} \\ 0 & \text{otherwise} \end{cases} \quad G_{ij} = \begin{cases} 1 & \text{treatment group} \\ 0 & \text{placebo group} \end{cases}$$

- Then the model is

$$Y_{ij} = \beta_0 + \beta_1 G_{ij} + \beta_{21} T_{ij2} + \beta_{22} T_{ij3} + \beta_{31} G_{ij} T_{ij2} + \beta_{32} G_{ij} T_{ij3} + \epsilon_{ij}$$

where $\boldsymbol{\beta_2} = (\beta_{21}, \beta_{22})$ are the '*Time*' coefficients and $\boldsymbol{\beta_3} = (\beta_{31}, \beta_{32})$ are the '*Group*Time*' coefficients.

Mean differences

- What is the impact of group/treatment when $j = 1$ (baseline), and $j = 3$?

- What is the difference between $j = 1$ (baseline) and $j = 3$ for the treatment group? How about for the control group?

- What is the difference in the difference between $j = 1$ (baseline) and $j = 3$ for the treatment and control groups?

Profile with interaction

When there is an interaction in a profile analysis, you cannot interpret:

- the effect of treatment without specifying the time point, and
- the effect of time without specifying the treatment group.

What you can do:

- give the difference in differences (commonly the main goal),
- discuss the impact of treatment separately by time point, or
- use least square means (LSMEANS) to get all the mean values and mean differences.

- If there is no group-by-time interaction, what β coefficients will be zero?
- Given there is no interaction, if there is no time effect, what β coefficients will be zero?
- Given there is no interaction, if there is no group effect, what β coefficients will be zero?

T-tests vs type III tests

- Most people focus on the individual t-tests when looking at the results of an analysis which will test, for example,

$$H_0 : \beta_{31} = 0$$

- This may be of interest, but commonly, we are interested in a group of coefficients all being equal to zero, for example,

$$H_0 : \beta_{31} = \beta_{32} = 0$$

  where type III tests should be used.
- **This is almost always true when using dummy coding.**
- In this situation, look at the type III test result first, then at the individual t-tests to see what is driving the results.

SAS Code

- To conduct a profile analysis of data from two or more treatment groups measured repeatedly over time, we can use the following SAS code.
- Which can be done with any kind of covariance matrix

```
proc mixed;
class id group time;
model y=group time group*time /s;
repeated time/type=UN subject=id r rcorr;
run;
```

GO TO SAS EXAMPLE

Summary

Profile Analysis:

- **Always** an option.
- Does not assume any specific trend.
- May have low power to detect specific trends, e.g., linear trends.
- You don't get an "overall" effect estimate.
- This is an option I like using for **exposures that are not of interest**.